

1 Quick Review

The idea of vector spaces has allowed us to provide an infinite amount of bases for almost any vector space (minus some wacky ones like $S = \{\emptyset\}$; try to think about why!). Now it's time to shift back to thinking about matrices (which are generally vectors) and the information we can extract from them.

Take some time to think about how long it may take for a chromebook to compute a fluid simulation with $500,000 \times 500,000$ elements (that's 2.5×10^{11} elements!). What if we found that we could represent the same system using just 10 vectors and 10 variables (the rest became zeroes!). We'd see a huge performance jump and we've just determined the **rank** of the matrix. The rank tells us how many variables and equations we actually need to describe a system. Think linear independence; if we had a set of 5 dependent vectors, do we *really* need all 5?

Mathematically (because that's what truly matters), the rank of a matrix is equal to the dimension of its row and column spaces.

Definition 1: Row Space

The row space of a matrix (represented by $\text{row}(A)$) is the space spanned by the row vectors of a matrix. A matrix with more rows (or equations) than the dimension of the row space is said to *overdetermined*.

Definition 2: Column Space

The column space of a matrix (represented by $\text{col}(A)$) is the space spanned by the column vectors of a matrix. A matrix with more columns (or variables) than the number of rows (or equations) in the column space is said to *underdetermined*.

Another key idea is the **null space** of a matrix. You've actually been introduced to this idea, but under a different name: the *kernel*.

A fun way to think about the kernel (and if you ever asked me how to describe it, you'll certainly have heard this because it's a fun explanation), is to imagine squeezing the entire universe into a single point (as shown below). The kernel is process used to do that.

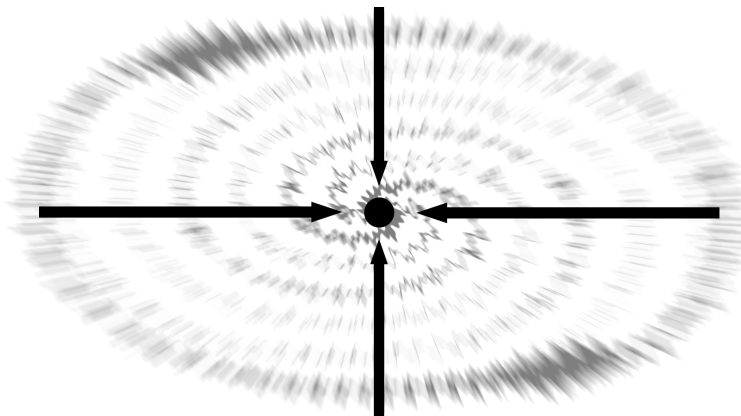


Figure 1: Honestly, I just wanted an excuse to make this.

2 Problems

1. The matrix $\mathbf{A}$ has been row-reduced as shown below.

$$\mathbf{A} = \begin{bmatrix} 2 & 7 & 0 & 2 \\ 2 & 0 & 0 & 6 \\ 5 & 3 & 0 & 6 \\ 1 & 0 & 10 & 3 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 10 & 3 \\ 0 & 1 & -10 & -3 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & -5 \end{bmatrix}$$

- (a) What is the row space of $\mathbf{A}$?
- (b) What is the column space of $\mathbf{A}$?
- (c) What is the null space of $\mathbf{A}$?
- (d) What is the rank of $\mathbf{A}$? _____
- (e) What is the nullity of $\mathbf{A}$? _____
2. Create a matrix $\mathbf{A}$ that has a nullity of 2 and a row space basis with 2 vectors.

3. **(PROOF)**. Prove that if an $n \times n$ matrix $\mathbf{A}$ can be row reduced to become the identity matrix I_n , then the row space takes the form of the standard basis of $\mathbb{R}^n$. Then, prove that $\text{col}(\mathbf{A}) = \text{row}(\mathbf{A})$.

4. Let $T_A : \mathbb{P}_2 \rightarrow \mathbb{P}_2$ be a transformation defined by $T_A(x_1, x_2, x_3) = (2x_1, x_3 + x_2, \frac{1}{2}x_2 + 5x_3)$.

(a) Find the null space of the transformation.

(b) If we were to replace x_1 from the image and replace it with x_2 , what would be the null space then?

- (c) Now let the vectors of T_A be a set of vectors. Is this set of vectors a basis for $\mathbb{P}_2$? Could it be a basis for $\mathbb{R}^3$?

5. Find the transformation matrix T that represents this sequence of transformations applied in $\mathbb{R}^2$:

- i) Rotation counterclockwise by 180° .
- ii) Shear up by 2.
- iii) Compression along the x-axis by a factor of 5.
- iv) Scale by a factor of 5.

6. Using T from question 5:

- (a) Prove that T is invertible.

(b) Find T^{-1}

(c) Apply the transformation T^{-1} to the vector $\vec{v} = (6, 7)$.